

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GET 210 — Past Questions & Materials
by Renance

Course: GET 210
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

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Paper 1 of 1 — GET 210 - 2024/2025 - Second Semester - ENGINEERING MATHEMATICS II

Level 200 • Second Semester • PastQuestion

COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF CHEMICAL ENGINEERING

Second Semester Examination, 2024/2025 Session

'COURSE TITLE: ENGINEERING MATHEMATICS II phere > yg :

COURSE CODE: GET 210 (GET 221) Time Allowed:

INSTRUCTIONS: Answer FIVE (5) questions only. Start each question on a fresh page. All questions carry equal marks. If you are taking this course as a Carryover/Legacy course, please enter

GET 221 in the "Course Code" row of your booklet.

(a) Given the relation defined by the set:

$$R = \{(1,2), (2,3), (3,4), (1,5), (4,4)\}$$

Is f a function? Explain your answer.

If f is not a function, make the minimal modification required for it to become one, and justify your modification. If R is already a function, simply state so - no justification is needed

(b) For the function s

on

$$x^3 - 27 \sqrt{x} + \dots$$

Find the domain.

«(Ww

i Simplify $\frac{1}{x^2}$ for $x \neq 3$. Note: 3 is not in the domain

ii. Calculate $\int \frac{1}{x^2} dx$, showing your work. Answer: $-\frac{1}{x} + C$

2. (a) Many practical problems in Engineering give rise to second-order differential equations of the form:

$$y'' + p(x)y' + q(x)y = r(x)$$

where p, q, r are continuous functions on an interval I .

7

Where a, b and c are constant coefficients and $f(x)$ is a given function of x . Show

$$y'' + 4y' + 3y = 0$$

When $a^2 + b^2 + c^2 = 0$ we

$$y = e^{ax}$$

: $b^2 + c^2 > 0$

Then show that $Y = Ae^{m_1 x} + Be^{m_2 x}$ which is a complementary solution with real and different roots to the auxiliary equation $m^2 + pm + q = 0$;

a}

' $a'y' + y = 13e^{2x}$, given that when $x = 0$, $y = 1$ and $y' = 0$

(b) Solve the equation, $y'' + 4y' + 3y = 13e^{2x}$, given that when $x = 0$, $y = 1$ and $y' = 0$

4

Oe 3. (a) Form a differential equation from $Y = Ax^? + Bx$. yt La dy

tetiers 1 :

(b) Solve the equation, oe $-5 - + 6y = 2\sin 4x$ 5 ie ly -) Rous the "

2 3 a

$= =$ and $= 3 \times ww) = -s$

: $d^?y d$; ae

(c) Solve the equation, ae $a > + 2y = x'$, given that when $x = 0$, a

4. (a) Interpret and solve the initial-value problem.

den ax

Saat lege t $2x = S$ cosat

$x(0) = 1/2$, $x'(0) = 0$

(b) Find the volume of the solid bounded by the planes $z = 9$, $X = 1$. Ray ae — surface $z = x + y$.

(c) Solve, $x =$ Byssy"

$dx \times 24 \ 2xy$ Fas

L oa)

5, If $A = (y^* - x^2z^2)i + (x^? + y^?)j - (x^?yz)k$, evaluate the following: (Bos $>$ a ca

(a) $\text{div } A$ at the point $(1, 3, -2)$ ot n a

oe

(b) $\text{Grad}(\text{div} A)$ at the point $(1, 3, -2)$ SA Oe ay we

(c) $\text{Curl grad}(\text{div} A)$ at the point $(1, 3, -2)$

oundary of

(d) Use Green's theorem to evaluate $I = \oint (4x + y)dx + (3x - 2y)dy$ where $\oint$ is the b the trapezium with vertices A (@,1), B 6,1), $\oint$ G3) D (1,3) Z (dec)

re 6. If $F(x) = x^7yi + 22Zj + 3z^2 xk$, evaluate $\int_C Fdr$ between A (0,0,0) and B (1,2,39 ; BR.

ao

(a) Along the straight line C1 from (0,0,0) to (1,0,0) , 2.2 oe

OF oe ae

(b) Along the straight line C2 from (1,0,0) to (1,2,0) :

\ VS

(c) Along the straight line C3 from (1,2,0) to G23) fs @, 7 .

(d) Along the straight line C4 joining (0,0,0) to (1233) ox "

7. (a) Show that $I = (3x^* \sin y + 2 \sin 2x)dx + (x^3 \cos y + 3xy^*)dy$ is independent of the path of integration, and evaluate if from A (0,0) to B $\oint$. Tt)

(b) Evaluate the integral $I = \oint xyds$ where $\oint$ is defined by the parametric equations

$= 3 = \sin^? - pe$;

$x = \cos^*t, y = \sin^*t$ from $t=0$ tot ; aed aN

gene

(c) Evaluate $\oint_C y \, dx$ when C is the circle $x^2 + y^2 = 4$ in the xy -plane.

C

; ; A

(d) If $F(\mathbf{r}) = x^2 \mathbf{i} + xz \mathbf{j} - 2yz \mathbf{k}$, determine $\oint_C F \cdot d\mathbf{r}$ over the curve C .

A

$F \cdot d\mathbf{r}$ between A ; ;

J from $(0,0,0)$ to $B(4,2,1)$ along the curve having parametric equations $x=4t, y=2t^2, z=t^3$ on

$x=4t, y=2t^2, z=t^3$; 1

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